

INTRODUCTION

AMIDST the artificial intelligence (AI) revolution, various user-centric applications have evolved to incorporate natural language processing (NLP) services to produce high-quality AI-generated contents (AIGC) [1]. This fusion allows machines to understand human languages, making AI more accessible and shaping a future where interactions with AI is as intuitive as human dialog. Traditionally, end users seeking access to NLP services have primarily relied on powerful NLP models located on cloud servers. Despite the high accuracy, cloud-based services may incur high latency and suffer from intermittent wireless connections, which makes it hard to maintain real-time guarantees [2]. A promising alternative to this is edge NLP, which facilitates NLP services on the edge servers. Thanks to the rich computational resources on edge servers and their proximity to user devices, end users can enjoy NLP services of both high quality and fast response [3], [4]. For edge NLP, the majority of works fall within the scope of distributed training, focusing on “how to train NLP models across edge servers to facilitate NLP services?” [12] In this work, we pivot our focus to a different approach by asking how to utilize trained NLP models (e.g., N-gram [5], BERT [7], and GPT [11]) deployed on edge servers for edge NLP? This approach is termed as distributed inference as we aim to exploit the inference capability of trained models across geographically distributed edge servers. For concreteness, as shown in Fig. 1, we study distributed inference for next-word prediction— a typical and non-trivial NLP service for end users. The goal of next-word prediction is to improve users’ input efficiency by proactively suggesting the most probable next-words based on inputs from mobile keyboards [13].